# Curriculum Vitae


## Md. Amit Hasan

Dhaka, Bangladesh

M.Sc. Oceanography | Researcher | Data Analyst


**Research focus:** environmental oceanography, water quality, coastal ecosystem health, remote sensing & GIS, machine learning, marine heatwaves, PFZ modelling, and blue economy studies.


### Contact

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**Website:** https://amithasanbd.github.io/


## Professional Profile

Early-career oceanography researcher with hands-on experience in environmental research, fieldwork, data analytics, geospatial applications, and technical communication. Currently in the final stage of an M.Sc. in Oceanography at Bangladesh Maritime University, with only the thesis semester remaining, and holding a B.Sc. in Oceanography. Demonstrates strong interest and growing expertise in marine and environmental science, remote sensing, machine learning applications, and research-based project support. Proficient in Python, R, ArcGIS Pro, Google Earth Engine, SPSS, and Power BI, with the ability to translate complex data into clear insights and organized reports. Aspires to pursue a fully funded PhD abroad in a relevant field while also pursuing part-time or full-time professional opportunities in research institutions, firms, NGOs, or development organizations to further strengthen academic, technical, and professional capacity.

## Education

**M.Sc. in Oceanography**
Bangladesh Maritime University | Expected Graduation: Dec 2026

**B.Sc. in Oceanography**
Bangladesh Maritime University | 2025 | CGPA: 3.55/4.00

**Higher Secondary Certificate (H.S.C.)**
Dhaka Board | 2019 | GPA: 4.67/5.00

**Secondary School Certificate (S.S.C.)**
Dhaka Board | 2017 | GPA: 4.91/5.00

## Core Competencies

- Remote sensing, GIS, and geospatial visualization
- Environmental data analysis with Python and R
- Google Earth Engine, ArcGIS Pro, QGIS, and SPSS
- Power BI, Excel, and analytical dashboard preparation
- Field survey support, coordination, and documentation
- Project reporting, presentations, and deliverables
- Research design support, literature review, and synthesis
- Machine learning for Earth and ocean applications

## Professional Experience

**Oct 2024 - Present** — **M.Sc. Student & Research Associate | Bangladesh Maritime University | Part-time**

- Contributed to ongoing research through data collection, data analysis, documentation, and technical reporting.
- Assisted in project summaries, presentations, organized outputs, and field monitoring,etc.

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| **Jun 2025 - Apr 2026** | **Assistant Research Officer \| DSD Research Firm, Dhaka, Bangladesh \| Part-time**<br>• Supported research assignments, project implementation, and preparation of reports and documentation.<br>• Coordinated field activities, survey logistics, data management, and team communication.<br>• Prepared presentations, summaries, and project outputs in deadline-driven settings. |
| **Dec 2024 - May 2025** | **Data Analyst \| DSD Research Firm, Dhaka, Bangladesh \| Contract**<br>• Processed and interpreted environmental and project data using Excel, Power BI, Python, R, ArcGIS Pro, QGIS, and SPSS.<br>• Prepared charts, analytical summaries, and visual outputs to support reporting and decision-making. |
| **Jan 2021 - Dec 2024** | **Researcher \| National Oceanographic and Maritime Institute (NOAMI),BD \| Part-time**<br>• Supported oceanographic and environmental studies through fieldwork, data collection, dataset processing, and report preparation.<br>• Worked with scientific instruments and multidisciplinary teams while maintaining organized research outputs. |

## Selected Research Projects

1. **Assessment of Existing and Potential MPA of Bangladesh** | *Jun 2023 - May 2024*

Assessed existing and potential marine protected areas to support biodiversity conservation, habitat protection, and sustainable marine resource management.

2. **Evaluation of the Status of Endangered Species using eDNA** | *Jun 2023 - May 2024*

Contributed to ecological assessment using environmental DNA techniques for species detection and monitoring.

3. **Impacts of Marine Heat Waves in the Coastal Region of Bangladesh** | *Jan 2023 - Dec 2024*

Examined temporal trends, environmental variability, and ecological implications of marine heat wave events.

4. **Ecosystem Health Assessment of Sundarbans Mangrove Forest of Bangladesh** | *Oct 2024 - Present*

Worked on ecosystem health evaluation focused on environmental condition, resilience, and conservation priorities.

5. **River Ecosystem Health Assessment of Dhaka City, Bangladesh** | *Jan 2025 - Dec 2025*

Assessed water quality condition, ecological stress, and river health dynamics in an urban environment.

6. **Assessing Climate-Induced Shifts in PFZs to Strengthen Coastal Resilience in Bangladesh** | *Jan 2026 - Present*

Linked oceanographic variability with potential fishing zone shifts and implications for coastal livelihood resilience.

## Publications and Manuscripts

### Peer-reviewed publications


1. Sowrav, S. F. F., Das, M. K., Debsarma, S. K., Anika, I. T., Hafiz, K. B., Hridita, N. T., Hoque, M. S. A., Broty, A. A., & Hasan, M. A. (2026). Water quality affects socioeconomic traits of climate vulnerable coastal people: case study on Dacope and Mongla. IDRiM Journal, 16(1). DOI: 10.5595/001c.158653
2. Hasan, Md. A., Koly, F. A., Sowrav, S. F. F., Rahman, S. M. M., Alam, N., Alam, M. M.-U., Kabir, Md. Z., Sarder, S. R., & Tonmoy, K. A. (2026). Assessing the dynamics of water quality and their implications on Bio-Chemical Health of river ecosystem: A RS-Based Study of the Shitalakshya River, Dhaka. Bangladesh Maritime Journal, 10(1), 313-329. DOI: 10.70279/bmj-v2-1124

### Selected manuscripts under review

- Predicting Spatial and Seasonal Variability of PFZ Using Satellite-Derived Oceanographic Data in the Bay of Bengal (Q1 journal, under review).
- Integrating Multi-Decadal Earth Observation and Community Evidence to Prioritize Mangrove Restoration in the Sundarbans, Bangladesh (Q1 journal, under review).
- Air Quality Fluctuations and Spatial Mapping Along a Major Transportation Corridor in Bangladesh: Implications for Public Health (Q1 journal, under review).
- Spatio-Temporal Mapping and Predicting the Distribution of Oxygen Minimum Zones (OMZs) in the Northern Bay of Bengal: A Remote Sensing and Machine Learning Approach (Q1 journal, under review).
- Emergence of a High-Exposure Marine Heatwave Regime in the Bay of Bengal (1995-2025): Intensification Patterns, Climate Teleconnections, and Ecosystem Impacts (Q1 journal, under review).
- Marine Spatial Planning and Zoning for Sustainable Blue Economy in Bangladesh's EEZ: Challenges and a Roadmap for Integrated Governance (Q3 journal, under review).

## Selected Conference Presentations

- Seaweed Cultivation and Its Potential for Sustainable Fuel Production in Bangladesh - poster presentation, International Seminar on UN-Decade of Ocean Science (2021-2030), Bangladesh Maritime University (2021).
- Effects of Climate Change on Ocean Acidification in the Bay of Bengal - oral presentation, International Conference on Climate Change, University of Dhaka (2022).
- Current Status of the Existing MPAs in the Bay of Bengal - poster presentation, International Conference on Oceanography, Bangladesh Oceanographic Research Institute (BORI) (2024).

## Awards & Recognition

- CCDB Climate Change Scholarship, 2025
- Third Place Winner, NOAMI Ocean Olympiad, 2025
- Best Trainee, Radar Meteorology Lecture Series, 2024
- Best Trainee (2nd), Data Science with Python Course

## Leadership & Service

- Founding President, Bangladesh Maritime University Ocean Club (BMUOC), May 2024 - May 2025
- Advisor, Bangladesh Maritime University Ocean Club (BMUOC), Mar 2026 - Present

## Technical Skills

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| **Programming** | Python, R |
| **GIS / Remote Sensing** | ArcGIS Pro, QGIS, Google Earth Engine |
| **Data / Research Tools** | SPSS, OriginPro, Power BI, Excel, data visualization, technical reporting |
| **Productivity** | MS Office, Google Workspace, Canva |
| **Research Methods** | Field research, survey support, data management, environmental assessment |

## Training & Certifications

| Training / Certification | Organization | Year |
|---|---|---|
| Marine Megafauna and Fisheries Survey Methodology | Healthy Ocean Project / World Conservation Society | 2025 |
| Certificate Course on Building Research Career for Early Career Professionals and Students | Nirmol Bangladesh | 2026 |
| Climate Change Scholarship with Training on Climate Literacy | Christian Commission for Development in Bangladesh (CCDB) | 2025 |
| Earth Observation for Enhanced Fisheries MCS | UNESCO/IOC | 2025 |
| Numerical Model & AI for Ocean Science (NMAI4OS) | NOAMI | 2024 |
| Ocean Model Configuration for OFS Implementation | IOC of UNESCO | 2023 |
| Data Science & Machine Learning with Python | NOAMI and Bangladesh Centre for Analytics | 2021 |

## Language Proficiency

| Language | Reading | Speaking | Writing | Overall Level |
|---|---|---|---|---|
| Bangla | ●●●●● | ●●●●● | ●●●●● | Native / Fluent |
| English | ●●●●● | ●●●●● | ●●●●● | Fluent |
| Arabic | ●●●○○ | ●○○○○ | ●○○○○ | Basic |

## References

| Dr. S. M. Mustafizur Rahman | Dr. Mohan Kumar Das | Sheikh Fahim Faysal Sowrav |
|---|---|---|
| Assistant Professor, Bangladesh Maritime University | Executive Director, Hydro-Climate & Ocean Center (HCO) | Chief Executive Officer, DSD Research Firm |
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